FINAL-YEAR PROJECT REVIEW | STUDY GUIDE

AutInsight

Technical walkthrough, research-paper plan, and viva preparation

What this guide is for
Use it to explain what is implemented, how the models and explanations work, what the results mean, and how to write a paper that accurately reflects the evidence. It separates the primary UCI label-reconstruction experiment from the secondary Indian cohort analysis.

Central scientific point

The UCI dataset class is exactly defined by the sum of the same ten questionnaire item scores supplied to the model: YES when the sum is at least 7. High model scores on this dataset therefore show reproduction of a questionnaire-derived label, not independent autism detection.

Defensible project description

AutInsight audits and models questionnaire-based pediatric ASD datasets, examines model behavior with SHAP, and presents a caregiver-oriented prototype. It does not establish a clinically validated screening or diagnostic system.

Keep this distinction visible in the paper, slide deck, and answers to reviewers.

1. Project in one view

AutInsight has a research pipeline and a web prototype. The research pipeline acquires and audits datasets, trains models, computes evaluation metrics and SHAP explanations, and saves artifacts and reports. The prototype lets a caregiver enter responses, receive a summary, and inspect an explanation of the model output.

Stage	Main operation	Output
Data acquisition	Retrieve UCI child screening data and the separate 2019 Indian S2 workbook; record provenance/checksums.	Source data and provenance manifests.
Audit and EDA	Inspect schemas, duplicates, missingness, target definitions, and distributions.	Audit reports, descriptive plots.
Modeling	Train Logistic Regression, Random Forest, and XGBoost on UCI; LR and RF on Indian cohort.	Saved metrics; primary XGBoost joblib artifact.
Explainability	Use TreeSHAP for local XGBoost output and global/local comparative RF analysis.	Per-case contributions; research SHAP summary.
Caregiver flow	Present adapted questions, context assessment, summary, next steps, and model explanation.	Local web interface using FastAPI inference.

The strongest contribution is an audit finding: the primary UCI target is score-derived and circular with respect to the questionnaire predictors. The model results should be reported as label-reproduction results, not as clinical detection results.

Scope distinctions

- The UCI cohort is the primary dataset for the deployed model artifact.
- The Indian S2 cohort is an independent, exploratory analysis with a different target and instrument; it is not pooled with UCI and is not the deployed model.
- The caregiver flow is not a validated AQ-10 administration: item text was paraphrased and four response options are collapsed to binary item scores.
- Counterfactuals, video analysis, calibrated reliability, and full explanation-quality validation are not implemented.

Project sources: `README.md`, `docs/methodology.md`, `docs/architecture.md`, and `docs/requirements_traceability.md`.

2. Datasets and target meaning

2.1 UCI child screening dataset

The UCI Machine Learning Repository child screening source has 292 raw rows and 21 columns (20 source fields plus target). There are two exact duplicate rows; training removes them before splitting, leaving n=290. The raw target counts are NO=151 and YES=141. Fields include A1-A10 item scores, age, gender, ethnicity, jaundice, family history, country of residence, prior app use, respondent relationship, result score, age band, and class.

The audit verifies for all source rows: `result = A1 + ... + A10`; `class = YES` if and only if `result >= 7`. `result`, constant `age_desc`, and identifiers/constant fields are excluded from the model inputs. However, A1-A10 remain inputs, and those items determine the target. This is target circularity, not accidental retention of the `result` column.

2.2 Indian AIIMS modified INDT-ASD cohort

The PLOS ONE 2019 supporting-information S2 workbook has 225 rows and 53 columns. Original target counts are 128 ASD and 97 non-ASD. One 180-month record exceeds the article stated 14-year maximum and is excluded for modeling, leaving n=224, 127 ASD and 97 non-ASD. The analyzed set uses 168 training and 56 test records.

The Indian model uses 32 features: 28 categorical instrument items, age in months, gender, SES, and residence. Exclusions include patient ID, old/new test-result summaries, section outcomes, onset/functioning summaries, severity, CARS, DP3, and DSM-5 target fields. Target is source code 1=ASD, 2=non-ASD, corresponding to expert-consensus diagnosis reported by the study.

2.3 Why the cohorts remain separate

UCI class is a questionnaire-score threshold; Indian class is expert consensus in a selected tertiary referral cohort. Instruments, target definitions, and populations differ. Pooling them or directly ranking their metrics would not answer a coherent prediction question. The Indian cohort is not community screening and its results do not validate the UCI model.

2.4 Data audit and provenance

Download scripts record source/licensing and checksums. UCI audit reports schema, duplicates, target consistency, and missingness. Indian audit reports schema, target counts, an out-of-range age, and leakage exclusions. Raw datasets are git-ignored. EDA outputs include age, class/target, missingness, and questionnaire response plots.

Key terms

Target: outcome label a model is trained to predict. **Target circularity**: the target is computed from predictors, so evaluation can reward reproducing that rule. **Leakage**: information unavailable at intended prediction time or an outcome proxy improperly enters predictors. **Spectrum/selection bias**: a referral cohort differs from the wider population where one might want to use a model.

3. Preprocessing and machine-learning models

3.1 Cleaning and transformations

- Trim whitespace and quotes in source strings; convert sentinels such as `?`, empty strings, `NA`, and `unknown` to missing values; normalize aliases in column names.
- Map UCI class YES/NO to 1/0. For Indian data, map target 1/2 to ASD/non-ASD and remove the out-of-range age row.
- Separate predictors and target; remove duplicate UCI rows; exclude `result`, constant age descriptor, IDs, diagnostic/target proxies, and other selected columns.
- Use a ColumnTransformer inside a scikit-learn Pipeline. Numeric branch: median imputation then StandardScaler. Categorical branch: most-frequent imputation then OneHotEncoder(handle_unknown="ignore").
- Split once with stratified 75/25 holdout and random_state=17. Fit preprocessing and model on training rows; apply the fitted pipeline to held-out rows.

The `StandardScaler` is useful for Logistic Regression. Tree models generally do not require scaling, but the current training script shares a preprocessing pipeline that applies it to numeric fields.

3.2 UCI models

Model	Configuration	Interpretation
Logistic Regression	max_iter=2000; class_weight=balanced; random_state=17; other sklearn defaults.	Linear baseline over transformed fields; models log-odds as a linear function of features.
Random Forest	400 trees; class_weight=balanced; min_samples_leaf=2; random_state=17.	Bagged tree ensemble; can model nonlinearities/interactions.
XGBoost	250 estimators; max_depth=3; learning_rate=.03; subsample=.85; colsample_bytree=.85; eval_metric=logloss; seed=17; n_jobs=1.	Gradient-boosted shallow trees; primary UCI runtime model and saved artifact.

There is no reported hyperparameter search. All binary metrics use a 0.5 probability cutoff. UCI LR and RF use balanced class weights; XGBoost does not specify a scale_pos_weight parameter.

3.3 Indian models

Only Logistic Regression and Random Forest are trained on Indian S2 data, with settings corresponding to the UCI baseline scripts (LR max_iter=2000/class_weight balanced, RF 400 trees/min_samples_leaf=2/class_weight balanced). There is no Indian XGBoost model in the current experiment.

3.4 Saved model artifact

`models/primary.joblib` stores the fitted XGBoost pipeline and associated explainer and metadata. The artifact manifest records dataset ID/checksum, target definition, training/test sizes, seed, software versions, XGBoost parameters, feature names, and artifact checksum. This lets the FastAPI runtime load the same fitted preprocessing and model used for prediction.

4. Evaluation protocol and results

The evaluation is one stratified 75/25 holdout, seed 17: UCI n=290 (217 train, 73 test); Indian n=224 (168 train, 56 test). There is no repeated or nested cross-validation, statistical significance test, or external validation.

Data/model	Accuracy	Balanced acc.	F1	ROC AUC	Brier
UCI Logistic Regression	0.973	0.974	0.972	1.000	0.019
UCI Random Forest	0.945	0.944	0.941	0.989	0.094
UCI XGBoost	0.973	0.973	0.971	0.998	0.037
UCI score rule baseline	1.000	1.000	1.000	-	-
UCI demographics-only LR	0.534	0.533	0.514	0.583	0.248
Indian Logistic Regression	0.857	0.849	0.879	0.945	0.088
Indian Random Forest	0.893	0.880	0.912	0.967	0.073

Interpretation: The direct UCI threshold rule has perfect agreement because it is the rule that creates the dataset label. High UCI ML scores are also mostly evidence of recovering this rule. The lower demographics-only result is a sensitivity analysis; it does not make the questionnaire-feature result diagnostic.

Metric definitions

Metric	What it measures	Key caveat
Accuracy	Share of test labels predicted correctly.	Can obscure class errors and target validity.
Balanced accuracy	Mean recall across classes.	Still evaluates the given target, not a clinical construct.
F1	Harmonic mean of positive-class precision and recall.	Depends on a selected positive class/cutoff.
ROC AUC	Ranking discrimination across thresholds.	Does not show calibration or clinical utility.
Brier score	Mean squared probability error against binary target.	A score is not proof probabilities are calibrated.

Do not compare UCI and Indian metrics as if they were a head-to-head contest. They use different labels and study populations. Report each separately and describe Indian results as exploratory.

5. Explainability: models, outputs, and limits

5.1 SHAP and Shapley-style feature attributions

SHAP (SHapley Additive exPlanations) allocates a model output among input features relative to a baseline. TreeSHAP is a tree-model-specific algorithm for computing Shapley-style attributions efficiently. In additive form: `model_output(x) = baseline + contribution_1 + ... + contribution_M`. A positive contribution moves the explained output up relative to the baseline; a negative one moves it down. Contributions describe the model, not causal mechanisms.

5.2 Primary XGBoost local explanation

- The saved XGBoost TreeExplainer operates on the preprocessed feature matrix.
- For each API prediction, the runtime builds a one-row frame in the artifact feature order, applies the saved preprocessing, computes SHAP values, sorts by absolute magnitude, and returns at most the top ten.
- The API labels the output scale `raw_model_score`. Thus the signed contribution is on the model output scale, not a percentage chance. The XGBoost additivity relationship has not been independently tested in this project.
- One-hot transformed feature names may be returned for categorical inputs. The UI maps recognized question/field names to caregiver-friendly labels where possible.
- The results page and detail view show horizontal bars. Bar widths are normalized relative to the largest absolute contribution in that individual result. This is a local contribution display, not a standard SHAP waterfall or beeswarm visualization.

5.3 Comparative Random Forest SHAP analysis

The UCI Random Forest analysis computes global mean absolute SHAP values across held-out records and one local held-out example. Global mean absolute attribution summarizes average model reliance in that test set, not global causal or clinical importance. For the one local RF example, baseline plus the positive-class SHAP values reconstructs the RF positive-class probability with absolute error 1.887379141862766e-15. This numerical check applies only to that RF model and record; it does not validate XGBoost explanations.

The top RF global attributions in the saved metrics are all A-item features, with A4, A9, A10, A5, and A8 at the top. That pattern is consistent with the score-derived UCI target. It is not evidence those behaviors are the most clinically important autism traits.

5.4 What is not implemented

- No SHAP explanation stability or perturbation-fidelity evaluation.
- No caregiver comprehension or clinician plausibility study.
- No counterfactual generator; response schema has an optional field, but no counterfactual method populates it.
- No project-generated TreeSHAP beeswarm or standard waterfall plot. The app’s local signed bars visualize returned contributions.
- No LIME implementation or systematic comparison among explanation methods.

Appropriate viva answer: “SHAP explains how a fitted model allocated its output across its input features. In UCI, those inputs also define the label, so the explanation describes the model’s reconstruction of a score rule. It cannot establish why a child has a behavior or a clinical diagnosis.”

6. Reliability, subgroup checks, and system implementation

6.1 Provisional probability tiers

The API defines confidence as max(p, 1-p) and assigns P0 at >=.90, P1 at >=.80, P2 at >=.70, and P3 below .70. P2/P3 are marked `deferred` by the API. These thresholds are not calibrated or clinically justified; no reliability diagram, threshold-selection study, JSD, or entropy method is implemented.

UCI model	P0/P1 coverage	Accepted test records	Observed accepted errors
Logistic Regression	89.0%	65/73	0
Random Forest	37.0%	27/73	0
XGBoost	75.3%	55/73	0

These zero observed errors are small single-split counts, not evidence that the thresholds are safe. There is also a front-end/backend distinction: API response has deferred status, but the current results page still reads and displays the positive/negative label rather than suppressing it. Therefore do not claim a complete user-facing abstention system.

6.2 Subgroup analysis

UCI reports gender/ethnicity subgroup counts, accuracy, sensitivity, and Wilson 95% intervals for sensitivity. Several ethnicity groups are very small. Indian outputs report gender subgroup counts and sensitivity intervals. These are descriptive uncertainty summaries, not a fairness guarantee or full fairness study. No formal parity criteria are established.

6.3 Application workflow and data handling

Part	Implementation detail
Frontend	Next.js 16 App Router, React 19, TypeScript; custom CSS. Routes: home, screening, results, results/understand. Recharts is a dependency for charts, while the new explanatory page uses static research figures and CSS contribution bars.
Questions	Ten caregiver-friendly AQ-10 Child-Version-informed paraphrases. Age gate is 4-11. Four agreement options collapse to binary item values; A1/A5/A7/A10 are agree-keyed, remaining items disagree-keyed.
Context assessment	Duration, settings, discussion areas, strengths, regression, prior support. It helps produce the caregiver summary but is not model-scored.
API request	Browser calls FastAPI POST /predict. The model row includes age and ten binary answers, with demographic/context values mostly fixed to unknown/default values in current UI.
Session storage	Caregiver observations and result are kept in tab sessionStorage; they are not persisted by an app database. The UI tells users not to include names or identifiers.
Backend	FastAPI, Pydantic schema validation, joblib-loaded pipeline. Endpoints: /health, /ready, /model, /research/metrics, /research/indian-metrics, /predict.
Deployment	Docker backend and Vercel frontend configuration exist. Project docs report no actual hosted deployment. Local XGBoost runtime on macOS requires OpenMP/libomp.

The model was trained using demographic/context fields, but current caregiver UI hardcodes gender, jaundice, and family history as unknown; defaults ethnicity/respondent/country to Unknown and app use to no. This must be disclosed as current inference-time behavior because it differs from collecting all model features from a caregiver.

Safety boundary

The adapted questions are not clinically validated; the app cannot diagnose or rule out ASD. It recommends professional discussion. There is no authentication, rate limiting, formal retention policy, or clinical governance. Do not enter real patient information.

7. Paper positioning and research questions

Suggested title

Auditing and Explaining Questionnaire-Derived Pediatric ASD Screening Labels: A Two-Cohort Machine-Learning Study and Caregiver-Facing Prototype

Suggested aim

To audit target construction and evaluate interpretable machine-learning models on a pediatric UCI ASD screening dataset, conduct a separate exploratory analysis on an Indian INDT-ASD validation cohort, and implement a caregiver-oriented prototype that communicates model behavior and limitations.

Research questions

- RQ1: How accurately do Logistic Regression, Random Forest, and XGBoost reproduce the UCI dataset class under a stratified holdout?
- RQ2: What does a deterministic audit show about UCI questionnaire item scores, total score, and class label?
- RQ3: How do LR and RF perform in a separate exploratory holdout analysis of the Indian modified INDT-ASD cohort?
- RQ4: What local/global model behavior can TreeSHAP expose, and what conclusions are not supported by those explanations?
- RQ5: How can outputs be presented to caregivers without representing them as a diagnosis or independent clinical probability?

Do not state that RQ3 externally validates the UCI model. It is a separate analysis against a different target and questionnaire.

Central contribution statement

This work contributes a reproducible audit of two distinct pediatric ASD datasets, documents exact score-label circularity in UCI, reports exploratory holdout model comparisons and SHAP analyses, and implements a caregiver-facing prototype that communicates the model's limits. It does not contribute a clinically validated ASD classifier.

Paper length

A reasonable target is about 5,500-6,500 words excluding references, tables, and captions, adjusted to the university or conference template. The outline below is a starting allocation, not a required publication format.

8. Research paper outline and writing checklist

Section	Words	Write this content
Abstract + keywords	200-250	Problem; objective; both datasets and their different targets; models; split; metrics; circularity finding; separate Indian result; prototype; conclusion and limits. Keywords: ASD screening, target validity, explainable AI, TreeSHAP, machine learning, pediatric cohort.
1. Introduction	600-750	Why screening support matters; screening vs diagnosis; dataset and label validity; explainability motivation; gap; aim, RQs, contributions. Do not imply current model is clinically usable.
2. Related work	700-900	Questionnaire ASD ML; age/developmental cohorts; clinical reference standards; calibration/deferral; explainable AI; Indian cohort research. Compare methods, validation, targets, and gaps. Verify every citation.
3. Data and targets	600-750	UCI source, 292 rows, fields, duplicates, missingness and exact score-rule target. Indian 2019 S2 source, 225 rows, out-of-range age exclusion, analyzed labels, selected referral setting. State no pooling/direct metric comparison.
4. Audit and preprocessing	400-550	String normalization, missing values, duplicate removal, target mappings, exclusions, numeric/categorical branches, train-only fitting, one-hot encoding, scaling, score leakage/circularity.
5. Models	500-650	LR, RF, XGBoost roles and exact parameters; shared sklearn pipeline; class weights; 0.5 cutoff; one split; primary artifact. Say Indian uses LR/RF only.
6. Evaluation	450-600	Stratified 75/25 split, seed 17, n per split; metric definitions; subgroup summaries and Wilson intervals; no repeated/nested CV or external validation.
7. XAI and reliability	500-700	SHAP additive idea; local/global TreeSHAP; RF additivity result scope; XGB local raw-score explanation; P0-P3 rule and limits; no JSD/entropy/counterfactual/quality metrics.
8. Prototype	300-450	Caregiver route flow; API; observations remain in tab storage; fixed demographic defaults; disclaimer; display of local contributions and next steps.
9. Results	550-750	Separate UCI and Indian tables; direct score-rule baseline; demographic-only sensitivity; subgroup and SHAP findings with correct model identity.
10. Discussion	600-800	Interpret UCI as label reconstruction; discuss separate Indian evidence; no significance claims; explain SHAP's contribution and boundaries; meaning for prototype.
11. Limits, ethics, future work	500-700	Single split, small samples, selection/target bias, no external validation/calibration, subgroup limits, adapted questionnaire, defaults, privacy/governance, no video/counterfactual.
12. Conclusion	150-250	Restate audit, exploratory model comparisons, XAI, prototype; say clinical validation remains future work.

8.1 Abstract recipe

- Sentence 1: establish the need to check target validity and explainability in questionnaire-based screening research.
- Sentence 2: state study aim and the distinct roles of UCI and Indian S2 cohorts.
- Sentence 3: state preprocessing, models, one stratified holdout, and metrics.
- Sentence 4: report exact UCI score-rule finding and model performance; report Indian metrics separately.
- Sentence 5: interpret UCI as label reconstruction and Indian result as exploratory.
- Sentence 6: conclude that the prototype is not clinically validated and further validation is required.

8.2 Results tables and figures

- Table 1: data cohort, target, inclusion/exclusion, and split counts.
- Table 2: model metrics with UCI and Indian results in visibly separate panels.
- Table 3: exact UCI score-rule baseline and demographics-only sensitivity analysis.
- Table 4: descriptive subgroup counts and intervals, flag small groups.
- Figure 1: data and application flow.
- Figure 2: audit graphic showing A1-A10 sum and threshold-to-class relationship.
- Figures 3-4: descriptive cohort plots and separate RF global/local SHAP outputs.
- Figure 5: an XGBoost local example from the app, labeled raw model-score attribution and not risk probability.

Do not create a combined score comparison that suggests UCI and Indian metrics measure the same outcome. Do not label SHAP importance as clinical feature importance.

9. Viva and review questions

What is the main finding?

The UCI class equals a threshold on the sum of A1-A10. Its model metrics primarily measure recovery of that score-derived rule, not independent ASD identification.

Why use ML when the direct rule exists?

The direct rule is a necessary baseline and audit. Comparing ML shows how models reproduce the dataset; the rule exposes that the task is already deterministic. This is a validity result, not a claim of clinical improvement.

Is this data leakage?

The explicit result score is excluded. The deeper issue is target circularity: the target class is generated from questionnaire items that remain model inputs. Describe both precisely.

What does XGBoost accuracy of 0.973 mean?

It matched 97.3% of UCI target labels on one 73-record holdout. It is not 97.3% ASD diagnostic accuracy.

Why include an Indian dataset?

It allows a separate exploratory analysis of the 2019 modified INDT-ASD cohort with expert-consensus target. It has different items and target and is not the deployed model or external validation of UCI.

How does preprocessing avoid leakage?

The split is made first, then the pipeline fits imputers, scaler, encoder, and estimator on training data only. The fitted pipeline transforms the test data.

What does SHAP mean here?

It allocates a model output to input-feature contributions relative to a baseline. It explains model behavior only; it cannot show causation or clinical reasons.

Why is TreeSHAP used?

The primary model is a tree ensemble. TreeSHAP computes SHAP-style attributions using tree structure efficiently.

Have you verified SHAP?

One comparative RF example has near-zero additivity reconstruction error. The primary XGBoost explanation has not been independently checked; explanation stability, fidelity, and user comprehension are also unevaluated.

Is the model calibrated?

No calibration model or calibration curve exists. Brier score is reported, but P0-P3 thresholds are provisional and uncalibrated.

Does the application abstain?

The API marks P2/P3 as deferred, but the current UI still displays the returned positive/negative label and does not enforce abstention. Full user-facing abstention is not implemented.

What do subgroup results prove?

They are descriptive counts, sensitivity and Wilson intervals. Small cells and one split cannot support a fairness guarantee.

What was the purpose of the caregiver interface?

To organize responses, show model behavior with explicit limits, and help caregivers prepare for professional discussion. It is not a clinical device or validated questionnaire.

What are the next steps?

Repeated/nested and external evaluation, calibration and decision analysis, full user-facing deferral, explanation-quality evaluation, appropriate data governance, and clinician/expert review. Counterfactual or video features require separately validated methods and oversight.

10. Claims to avoid and core terms

Claims to avoid

- “The model diagnoses autism.”
- “The model has 97% ASD detection accuracy.”
- “The predicted probability is the child’s chance of autism.”
- “SHAP explains why the child behaves this way.”
- “The model is fair” or “generalizes across populations.”
- “The Indian dataset validates the UCI model.”
- “The app has a clinically validated abstention system.”
- “Counterfactuals or video analysis are implemented.”

Terms to use accurately

Term	Meaning in this project
Screening class	UCI dataset label produced by questionnaire score threshold; not clinical diagnosis.
Probability	Model’s estimated probability for the dataset class; not clinical risk in this prototype.
Local explanation	Attributions for one model output/input.
Global importance	Aggregate model reliance across a dataset/sample; not clinical feature importance.
Calibration	Agreement between predicted probabilities and observed event frequencies; not implemented here.
Selective coverage	Share of records accepted by a deferral rule; computed exploratorily for P0/P1.
Counterfactual	A minimally changed input that changes model output, subject to feasibility/plausibility rules; not implemented.
External validation	Evaluation on independent data beyond the training study; not available.

Most useful project files

Start with `docs/methodology.md`, `docs/model_card.md`, `docs/data_dictionary.md`, `docs/requirements_traceability.md`, `scripts/train_models.py`, `scripts/train_indian_models.py`, `models/artifact_manifest.json`, the two files in `experiments/results/`, `backend/app/main.py`, and `backend/app/services/runtime.py`. The caregiver flow is in `frontend/src/app/screening/page.tsx`; question wording/scoring is in `frontend/src/lib/questionnaire.ts`.

References to verify and cite in your final paper

UCI Machine Learning Repository, Autism Screening on Children (dataset 419), DOI 10.24432/C5659W. PLOS ONE (2019), Development and validation of a DSM-5 based diagnostic tool for children with Autism Spectrum Disorder, DOI 10.1371/journal.pone.0213242. Lundberg and Lee (2017), A Unified Approach to Interpreting Model Predictions, NeurIPS. Use the AQ-10 Child Version source guide for instrument background and cite the exact public source in your style. Verify the bibliography and source claims for every additional literature-review reference before submission.

Final defense reminder

The most credible answer is often a scoped one: state what the experiment actually measured, identify its limitations, and distinguish a research prototype from clinical evidence. A candid target-validity finding is stronger than overstating a performance score.